



Abdiel M. Vallejo Padró

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OBJECTIVE

Seeking an opportunity to be part of a strong, multitasking, and leadership oriented team. Pursuing an Internship or CO-OP where I can apply and maximize my talents, including knowledge and experience in design, analysis, and other relevant technical projects.

EDUCATION

B.S. Computer Engineering, University of Puerto Rico Mayagüez Campus Expected: 2029
General GPA: 3.53

TECHNICAL EXPERIENCE

SPARC, UPRM Aug 2024 - Present

- Developing current state of the art Machine.
- Working with partners by using NASA technology.
- Understanding of research and papers to promote new ideas
- Research for new ways to improve the performance of AI in lunar rovers.

NASA RASC-AL, Competition June 2025

Lead Hardware and Autonomous systems Engineer

- Led hardware & autonomy development for a lunar rover, integrating sensing, power, computing, and mobility subsystems under extreme lunar constraints.
- Developed an autonomous navigation stack and fault-tolerant autonomy framework based on SHERPA-style redundancy, enhancing system resilience to radiation events and software faults.
- Collaborated with mechanical and robotics teams to improve robotic-arm stability, reduce power draw, and enhance payload handling performance.
- Led and directed the development of a ROS Noetic + Gazebo simulation (Ubuntu OS) to evaluate optimal sensor placement, verify perception coverage, and analyze lunar surface illumination conditions for improved autonomous navigation.

VOLUNTEER EXPERIENCE

Robotics Pro Competition, NASA booth Oct 2025

- Presented our NASA RASC-AL project to adults and children at a public outreach booth.
- Explained the project architecture, objectives, and innovation to a diverse audience.
- Adapted technical explanations to different knowledge levels.
- Represented the team in a professional STEM outreach environment.

AI Winter School Jan 2025

- In-depth AI training led by industry professionals.
- Hands-on Python development using Jupyter Notebook, Google Colab, and VS Code.
- Practical application of AI concepts in guided exercises.

TECHNICAL SKILLS

- Proficiency in Microsoft 365
 - Proficient in Python
 - Proficient in C
 - Proficient with Linux
 - Proficient with Cybersecurity tools
- Proficient in hardware and computer assembly
 - Proficient in Git
 - Beginner in MATLAB
 - Beginner in Gazebo simulations
 - Beginner in SolidWorks